AI-Powered Smart Contracts: The Dawn of Web 4. O

Iraq Ahmad Reshi, Muneeb Zahoor Khan, Sadaf Shafi, Sahil Sholla, Assif Assad, Huzaib Shafi

***Abstract*— Smart contracts backed by artificial intelligence hold the key to a better tomorrow. As smart contracts begin to incorporate artificial intelligence, many previously steady marketplaces might instantly turn unpredictable. A smart contract is a program that operates autonomously on a blockchain. The program's decentralization ensures its safety, openness, and credibility. However, since they are static in essence and cannot adapt to new circumstances due to their immutable nature, these programs are limited in the tasks they perform. As a remedy for this shortcoming, we propose a few novel use cases where we solve some of the huge problems in a wide array of domains by incorporating Artificial Intelligence in these programs called Smart Contract. With consideration of hardware and new frameworks, this manuscript also discusses the potential of utilizing AI in such contracts. We describe the usage of AI to monitor the activity of these smart contracts in real-time and to provide an alert if an attack seems imminent. We also offer a new approach for marketing businesses and organizations on the blockchain (DAO) that is likewise decentralized and makes use of AI in the smart contract to enable incentivization. The use of AI in Smart Contracts also opens the door to the possibility of other applications, such as Fashion Design in Gaming.**

***Index Terms*— Anomaly Detection, Artificial Intelligence, Blockchain, Fashion Designing, Smart Contracts,**

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I. Introduction

A smart Contract can be defined as an automated protocol that gets executed after pre-determined conditions are met. Typically, smart contracts are used to regulate the implementation of an accord so that all parties may know the conclusion instantly, without the need for a mediator. In addition, they may automate a process by initiating the next operation when certain circumstances are satisfied [1]. Smart Contracts, which exhibited major growth with blockchain, have attained considerable value presently, despite their origins in the 1990s. Research on smart contracts has exploded in popularity in recent years. In principle, smart contracts incorporated in Blockchain technology enable contract conditions to be autonomously handled without the participation of an intermediary. The use of smart contracts is revolutionizing the commercial world. Blockchain-based smart contracts automatically enforce agreed-upon terms, eliminating the need for a neutral third party to mediate disputes [2]. By streamlining administrative costs and boosting the effectiveness of company processes, smart contracts can help mitigate risks.

The concept of a smart contract relies on the translation of agreements into code that can be shared between various nodes for verification purposes. However, contemporary smart contracts are more akin to digital contracts than true smart ones. Real-world distributed programs on the blockchain are challenging to develop without better smart contract capabilities [3]. Due to its inability to facilitate dynamic decision-making and the immutability of smart contracts, blockchain technology has limited its practical commercial uses. The integrity of the projected result cannot be maintained by artificial intelligence, a computer prediction platform that provides intelligent forecasts, operations, and identification. Nevertheless, with the present state of these techniques, it is still not possible to guarantee concerns like the presence of consenting faults, flaws in the ability of the contractual parties, financial regulation, or the manipulation of such contracts for illegal reasons. Contract comprehension issues, the rigidity of code language, enforcing court judgments while taking into consideration the blockchain's immutability, and the operation of so-called oracles—the points where the physical and virtual worlds considerably overlap along with other difficulties need to be addressed. To create a decentralized machine learning architecture with enhanced security, reliability, and adaptability, blockchain-based AI projections can fill in the gaps left by existing technologies [3,4].

Smart contracts could be a fully functional convention having complete legitimacy in the actual world, provided AI is used to improve them[5]. Figure 1 depicts the scenario where Artificial Intelligence powered smart contracts are deployed on a blockchain network in its off-chain mode. AI-trained models have proven efficacy in pattern recognition and matching techniques. The deployment of a contract is shown, which is subjected to a pattern matching algorithm. The deployment fails in case the matching results obtained by the algorithm are adverse. The AI operations usually occur off-chain as the on-chain deployment of AI models is cumbersome [6]. The inclusion of artificial intelligence in smart contracts increases their productivity by a significant margin [7]. There is an amazing improvement in the efficiency of smart contract processes, particularly with Blockchains. The removal of human interaction in contract validation accounts for the bulk of this efficacy. As a result, the negotiating process is simplified and streamlined. The arrival of AI in the field of smart contracts ushers us into a new age, one that will see many enterprises and legal experts succeed. By rendering Smart Contracts highly adaptable, AI can improve their efficacy. Logic, neural graphs, and neural networks are all examples of adaptive systems. The AI can create and execute Smart Contracts based on essential analysis, which means that AI will predict whether to execute a smart contract or not [8]. Moreover, increasing growth of NLP is necessary for AI to realize its capabilities since it can analyze not just smart contracts but also "conventional" contracts, relevant laws, and court rulings, as well as the intentions of the participants. The merger of the two techno giants guarantees three A's (AAA) of digitalization (IBM). An audit trail is created when AI models are stored and distributed via blockchain, and combining Blockchain with AI can strengthen information integrity (Authenticity). AI is capable of reading, comprehending, and correlating data quickly and thoroughly. Blockchain enables AI to evolve by allowing access to massive amounts of data from both inside and outside the organization, managing data consumption, and model distribution (Augmentation). Blockchain-based smart contracts with AI models incorporated make recommendations, carries out transactions and maintains integrity checks (Automation)[9]. Despite all these numerous pros, embedding AI in smart contracts in its on-chain mode needs a lot of assessment as training AI models on blockchain is cumbersome[3,5,6,8]. Some of the frameworks that have been designed while integrating blockchain and Artificial Intelligence are mentioned in next section. Section 3 briefs us about the challenges while incorporating AI in blockchains. In Section 4 we propose novel use cases that incorporate AI in Smart contracts and is followed by a conclusion section.

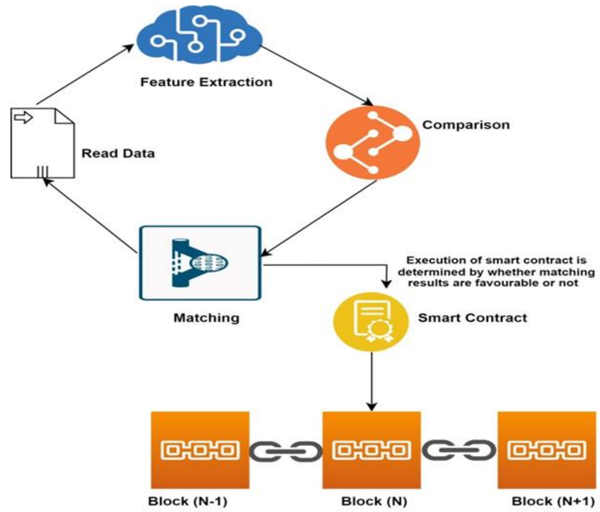

Figure 1 AI-driven Smart Contract Deployment in Blockchain.

II. Platforms with AI-Based Smart Contracts

A different set of prospects for a variety of industries has opened up by the development of blockchain and its fusion with Artificial intelligence and machine learning. Prospective blockchain initiatives may frequently involve using AI to solve blockchain trilemma [10]. Furthermore, it appears that blockchain offers multiple opportunities for current AI models to be improved, such as by empowering marketplaces to cut the operating costs for AI services or by assisting both humans and AI in understanding the reasoning behind a given conclusion reached by the AI engine. By assigning rules and policies to the chain, AI can be incorporated in smart contracts, resulting in the merging of smart contracts with self-learning artificial intelligence. AI is better at predicting outcomes when it has access to more data [11]. AI might, for instance, function as a recommendation system in supply chain contract negotiations as it can see how participants engaged before by examining recorded contracts. Then, AI may review earlier contracts to find aspects that were omitted and include them in forthcoming negotiations. Smart contract cost and rules and policies are the two important parameters, according to Almasoud et al., in [11] that need to be considered while designing AI-enabled smart contracts.

AI and Blockchains are emerging as an alliance to create an amazing and better future. The collaboration is multidimensional and seeks to contribute to all sectors. Figure 2 depicts various platforms that emerged from the alliance of AI and Blockchain. One of the most important aspects of all developed frameworks is that each framework is designed to solve problems of a specific type. However, no AI-based blockchain has been developed yet that can serve as a criterion for deploying AI-based smart contracts.

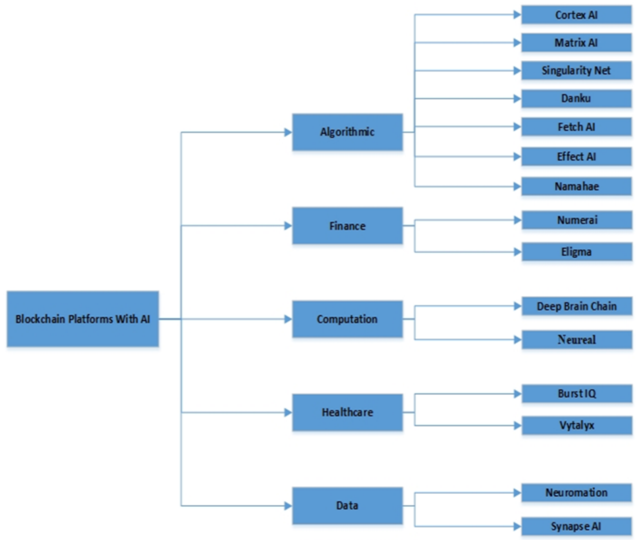

Figure 2 Blockchain Platforms utilizing AI for various services

A. Algorithmic

This section briefs about certain blockchain projects that incorporate AI algorithms in blockchain to solve certain specific challenges.

Cortex currently supports AI algorithms in smart contracts, allowing anybody to incorporate AI into smart contracts [12]. At the same time, Cortex has developed a collective cooperation incentive system that allows anybody to contribute and optimize designs in Cortex, with contributions to the models being rewarded. AI decentralized applications (DApps) attain consensus on-chain after being run on each complete Node employing GPUs and FPGAs in Cortex. Cortex is poised to generate numerous models that surpass human capabilities, similar to what is happening in other domains, given its openness and sharing aspects. The core concept is blockchain, wherein users can add Cortex's suggested AI system to existing smart contracts. Such cooperative effort potentially results in the development of Artificial General Intelligence on the Cortex framework, enabling decentralized AI applications to be quite reliable yet complicated. Cortex features a built-in emulator that is able to run Ethereum smart contracts and is interoperable with Ethereum Virtual Machine (EVM). Users can utilize the Cortex virtual machine by paying Cortex assets to employ AI algorithms which are a part of the Cortex infrastructure.

MATRIX's development as an intelligent blockchain is too aided by artificial intelligence. The objective of MATRIX is to create an intelligent network that can fully use the capabilities of blockchain technology. Moreover, MATRIX is the result of substantial blockchain technology improvement and expansion. Owners of smart contracts no longer have to know how to code in programming, which is a significant advantage of MATRIX [13]. The fundamental transaction sequences will be identified and converted into a program that captures the behavior of the targeted smart contract via a deep neural network-based code generator. By offering some additional components like a rule-based semantic and syntactic analytic engine for smart contracts, a verification toolset to give the security features of a smart contract, an AI-based algorithm that finds issues in transaction models and assesses its security, and finally, a deep learning platform for comprehensive security verification and enhancement Matrix overcomes the challenges associated with smart contracts. Over traditional blockchain's matrix possesses automatic smart contract generation, secured smart contracts, and value-added mining that can be unleashed for real-world applications.

SingularityNET is a blockchain-based platform for AI systems to connect with one another and with external consumers [14]. Developers may publish their AI tools on the network, where they will be able to interact with other AIs as well as paying consumers. Contracts for execution by external, non-AI Agents that desire to get AI services from Agents in the community are included in SingularityNET's smart contract repository. Anyone may develop a node (an AI Agent), make it public (on a server, workstation, drone, or embedded device), and connect it to the channel so it can demand and/or accomplish AI tasks while interacting with other modules and conducting economic transactions.

DanKu is a novel Ethereum-based blockchain-based platform for assessing and purchasing machine learning models developed by Algorithmica [15]. DanKu makes high-quality, objectively verifiable machine learning models available to anybody. The DanKu protocol makes use of smart contracts and blockchain technologies. Anyone may publish a large dataset, an assessment function, and a monetary incentive for whoever can give the best powerful machine learning model for the data within the terms of the contract. The contract enables the establishment of a decentralized and trustless marketplace for the exchange of machine learning models. This allows machine learning practitioners to directly commercialize their talents.

Namahe is a collaboration of AI and Blockchain where aggregation of all stakeholders occurs on a single platform. This ensures more transparency as the tier system vanishes, which results in reduced child labor and ensures workers are paid sufficient minimum wages[16]. AI acts as a pre-emptive tracing agent and predicts the issues inertly in advance so that mitigation strategies can be formulated. Built on hyper ledger sawtooth platform, the access-controlled namahe aims to revolutionize supply chain management by utilizing smart contracts and AI.

Fetch.AI is an inter-chain mechanism called Fetch.ai Blockchain employs a powerful smart contract (Cosmwasm). It enables the implementation of cryptography and AI logic on the chain. Useful Proof-of-Work(UPoW), a distinct consensus technique, is employed by Fetch.ai[17]. The amount of work completed in between the production of two blocks determines the sequence of transactions in UPoW. Without disclosing the original information to either of the participants, scattered entities can collaborate to train ml models using the Fetch.AI collaborative learning module. It enables and trains its net to learn from confidential devices without having accessibility to it by combining blockchain technology with AI learning ability.

Effect.AI launched in 2017 with an aim to provide frictionless entry into web3.0. Effect.AI suggests a free, open, and distributed platform that offers the majority, though not all, of the services necessary for a competitive artificially intelligent product[18]. By launching the initiative across NEO blockchain-based platforms that are powered by the network token EFX, Effect. AI hopes to achieve its goal. Effect.AI is seeking to undermine the Amazon Mechanical Turk (developers require humans to train their algorithms for a certain activity) business by employing smart contracts set up on Neo blockchain technology to take out the middlemen and significantly reduce the unnecessary computational and operational costs in it.

B. Finance

Numerai is an AI-based hedge fund platform maintained by data scientists. Computer scientists can participate in Numerai's weekly tournament and submit their estimates for a chance to win prizes given out in the Numeraire cryptocurrency[19]. In order to forecast the financial markets, you can create machine learning models using abstracted financial data in the Numerai Tournament. With NMR cryptocurrency, you may stake your models to get performance-based incentives. In 2016, Numerai secured $7.5 million in two rounds of investment.

Eligma is an AI-driven, blockchain-based marketplace that is intended to change the entire traditional marketing[20]. Eligma allows users to buy, sell and track their assets. The working algorithm of eligma is built on two cores, an AI chatbot for user preferences and value prediction and blockchain core for decentralized sales point. Value prediction algorithm on AI core focuses on future, and current estimation of products and the decentralized sales point is a site where a smart contract binds the interacting parties.

C. Computation

DeepBrain Chain (DBC) is an artificial intelligence (AI) computing platform built on the blockchain with the goal of reducing processing costs for users. It uses the notion of ANNs in a distributed manner across N nodes by use of blockchain technology[21]. Using Polkadot Substrates, DeepBrainChain creates its primary network. The system offers access to a distributed network harboring vast quantities of computational power. DeepBrain Chain's ultimate goal is to be the most massive computational platform in the AI+ metaverse age by creating a decentralized powerful computational network with limitless scalability[22].

The objective of Neureal is to remove the expense of high performance computing and the inability to manage large amounts of data, especially big data dumps, enabling everyone, from large organisations to lone individuals, to always be accurate [23]. The Neureal Network makes it possible to commoditize idle computer capacity and use it for big data analysis and future predictions while awarding miners for their accurate forecasts. Each element of Neureal is constructed in a modular way to allow for easy repair as needed. The stacking architecture of the built-in mining software and its classifiers, which enables the output of one prediction to be the input of another predictor, is a significant element in Neural.

D. Healthcare

The BurstIQ platform offers the technical backbone necessary to simplify the process of collecting, evaluating, and analyzing data for millions of people over extended durations. Conventional blockchains' scalability and throughput issues are solved by BurstIQ, which replaces the proof-of-work process with a voting model among trustworthy nodes and a consensus mechanism at the persistence layer[24]. LifeGraphs are constructed utilizing a machine learning capacity that generates a relational map of verified data items (in this case, a person's health records). Individuals' health records and the parameters under which they may be accessed are protected by smart contracts called consent contracts.

A paradigm change in healthcare and wellness decision-making is what Vytalyx seeks to achieve as the best-in-class interoperable and AI-driven blockchain technology. In order to create AI-driven Personal Health Information, Vytalyx integrates an unprecedented amount of clinical data (such as anonymized EMR/EHR data, individual medical histories, clinical research, and biomarkers, including genome and microbiome data), as well as non-clinical data (such as wellness, portable, real-time device, nutrition, and fitness data)[25]. Due to its potential flexibility, adaptability, ability to interface with various public, private, and cross-chains, and capacity to manage higher quantities of data on-chain, Vytalyx leverages Hyperledger's Sawtooth Burrow, which uses Proof of Elapsed Time ("PoET") as its consensus mechanism.

E. Data

The Neuromation Ecosystem may transform AI model creation using distributed computing and blockchain proof-of-work tokens. It includes in a single location the key components for developing deep learning algorithms using synthetic data. Nodes can run Neuromation Computation Node in addition to the regular mining program[26]. If Node wins the auction, it will redirect its computer resources from mining Ether or another cryptocurrency to work on the Neuromation platform. The Node will produce synthetic data or train a model using Deep Learning. The miner will be rewarded with Neurotokens. After the job has been completed, the Neuromation Node will quit, and the miner will continue mining cryptocurrency.

Synapse AI is a network that pays users with SYN tokens (a cryptocurrency) in exchange for sharing their personal data. SYN utility tokens, which can be exchanged for other major exchanges globally or utilized for services, are gained as cryptocurrency incentives[27]. SynapseAI and others can train machine learning models using user-submitted data. With SynapseAI, users may select the data they share and the source device for their data.

It is becoming increasingly vital to digitize as many functions as needed in a society wherein human factors is gradually becoming the primary cause of mistakes, from minor ones like forgotten medicine to large ones like Chernobyl disaster. Creating automated organizations would be the first positive step. Authors in [28] created an art group that makes money by selling paintings. The goal of this project is to construct a GAN that makes art from online scraped photos, emulating the work of great painters. The portal from which a user may purchase these artworks will retrieve these photos. The costs and quality will be determined by the decision-making Artificial Intelligence that will serve as the organization's brain. Blockchain and AI work in collaboration in the mentioned project as it consists of the following modules. GAN Employee to produce images upon training. Website serves as a front-end interface for purchasing. Smart contracts through Remix and Metamask to download the content and make payments. Decentralized computing infrastructures like Golem and iExec are proposed for this project to achieve fully decentralized infrastructure. Decision-making AI makes decisions, including how to share earnings, which artwork categories to generate, which GAN framework provides the best-selling images, and so on.

Apart from Cortex, no existing framework comes even close to incorporating AI in smart contracts. AI models in Cortex are uploaded into the blockchain that later can be integrated with smart contracts. The Cuckoo Cycle method was chosen as the consensus mechanism because it can be mined using modest equipment. In spite of being a memory-intensive algorithm, Cuckoo Cycle has the advantage of being instantly verifiable, making it one of the most intriguing ASIC-resistant frameworks. For deterministic AI inference, Cortex offers "Synapse," an integer inference engine. Synapse Technology, developed by Cortex, is a deterministic inference engine that ensures an AI model always yields the same result, even when deployed in a heterogeneous computing environment. Synapse uses quantization of AI models in conjunction with deterministic GPU acceleration to provide access to AI-powered decentralized applications. Using quantization, models may be implemented on the blockchain at little cost, and consistent inferred outcomes can be obtained. The blockchain and associated capabilities are, however, continuously evaluated and need to be enhanced accordingly. Currently, research to develop a perfect, fully automated, and thorough MRT (Model Representation Tool) for AI-based Smart contracts.

Table 1 Depicts Various AI-based Smart contracts initiatives

Platform	Contribution	Reference	Consensus Mechanism
Cortex	Augmentation of smart contracts by AI and allowing anybody to incorporate AI into smart contracts	[12]	PoW
MATRIX	A Neural Network using, delegated PoW, rules attached to the contracts, and automated conversion to transform simple scripts into SCs.	[13]	PoW and PoS
Eligma	Single market that employs AI services to increase transparency and proactively spot issues	[20]	PoW
SingularityNET	Integration of different AI services and marketplace where algorithm developers may sell their work.	[14]	PoW
DanKu	Establishment of a decentralized and trustless marketplace for the exchange of machine learning models	[15]	PoL
Namahe	Transparent and secure supply chain	[16]	BFT
Fetch.AI	Implementation of cryptographic protocols on chain	[17]	UPoW
Effect.AI	Develop a BC based platform to simulate AI algorithms	[18]	DBFT
Numerai	AI based hedge fund platform	[19]	PoS
Neural	Blockchain-based open source, peer-to-peer AI supercomputing	[23]	Hybrid Consensus
Vytalyx	Vytalyx intends to store and analyse medical data using blockchain and AI.	[25]	PoET
Synapse AI	Worldwide data marketplace that is decentralized and built on the Ethereum blockchain.	[27]	PoW

III. USE CASE ILLUSTRATIONS FOR CONVERGENCE OF AI IN SMART CONTRACTS

A. AI for anomaly detection of blockchain data

Blockchains and smart contracts cannot inherently access data that resides outside the blockchain environment. However, accessibility to the data about real-world state and events is vital for a lot of smart contract use cases, like DeFi, insurance, gaming and NFTs, etc.[29]. Herein lies the value of blockchain oracles. Oracles are third-party services that act as intermediaries between Blockchain and external systems, allowing smart contracts to consume off-chain data and make decisions/actions accordingly. The blockchain oracle itself is not the data source; rather, it is the layer that queries, verifies, and authenticates external data sources and then relays it to the blockchain environment[30]. Since the outcome of smart contracts directly depends on the data delivered by oracles, it is highly critical to ensure the reliability and accuracy of this data. Faulty data can lead to smart contracts executing the wrong outcomes. As blockchain transactions are automatic and immutable, a smart contract outcome based on incorrect data is irreversible, leading to a permanent loss of user funds [31]. As anyone can operate a blockchain node in a pseudo-anonymous manner and submit responses, it is incredibly challenging to handle and implement quality for off-chain data. As oracles deal with data that is non-deterministic, hence reliability check needs to be done. Users' loyalty in the blockchain, which view it as more dependable than conventional systems, might be jeopardized by Oracle problems when deployed via smart contracts. Even if the oracle is reliable and untouchable, there is always a potential that the information it is processing has tampered. In such a case, despite being a reliable device, it will send false information to the smart contracts. Chainlink made a significant effort to contain the oracle problem. To mimic the
35 blockchain's consensus algorithm, the start-up developed a system of decentralized oracles based on reputation. This robust system effectively handles Oracle malfunctions or failures, although service providers might still engage in collusion or purposeful data manipulation. Research indicates Chainlink is agnostic since it is still in its early stages[31]. Chainlink relies on centralized verification and conflict resolution since it lacks a reliable method of determining whether the information sent to you is valid or not. Moreover, requestors are unable to confidently vouch for the correctness of the inputs and the financial impact of providing incorrect information, while node suppliers are unable to appropriately compensate for errors[29][30]. Chainlink increases the attack surface. A cyberattack on Chainlink hubs costs them at least 700 ETH [31]. When decentralization is insufficient to solve the oracle problem, and it is impossible to objectively verify the consistency of the results, the smart contract ecosystem needs a "trust model" to maintain some level of dependability.

This is where the integration of AI and Blockchain could prove potentially promising. Many AI techniques could be used to ensure a facet of data quality in terms of anomaly detection. Anomaly detection is the process of identifying irregular patterns, aberrant behaviors, or outliers of a data set. Simply put, it involves finding patterns that do not conform to expected behavior. These could be indicative of problems such as device/sensor malfunction, intrusion attempts, fraudulent transactions, and so on[32].

In some cases, anomalies could simply be genuine infrequent occurrences. An analysis of anomalies can assist in understanding the context, eliminating possible causes (if necessary), and improving data quality. Anomaly detection has been a well-studied area for a long time. Many AI techniques have been proposed in this regard, including Density-based approaches (such as Robust Kernel Density Estimation, Ensemble Gaussian Mixture Model), Quantile-based methods(such as One-class SVM, Support Vector Data Description), Neighbor-based methods(such as Local Outlier Factor, kNN Angle-Based Outlier Detector), and Projection-Based methods(such as Isolation Forest, Lightweight Online Detector of Anomalies)[33]. In addition, there are also some deep learning-based approaches for anomaly detection, such as autoencoders, sequence-to-sequence models, generative adversarial networks (GANs), and variational autoencoders (VAEs)[34].

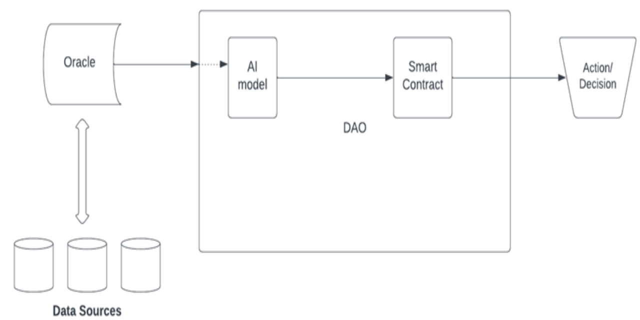

Figure 3 AI-enabled Smart Contract for Anamoly Detection.

In the context of blockchain, an anomaly detection model can be used at the oracle level, where the model will aid in detecting anomalous behavior in the data aggregated by the oracles from different sources. The oracle network can then inspect the detected anomalies and perform any action necessitated before proceeding further with the data. This would ensure that any aberrant behavior in the data is addressed beforehand, preventing any adverse impact on the smart contracts' outcomes consuming this data. To demonstrate this use case, we will use the Isolation Forest (IF) model to detect anomalies in a dataset. Isolation Forest [35] is an ensemble regressor, and it uses the concept of isolation to "explain/separate out" anomalies. The underlying idea is that it is relatively more straightforward to isolate anomalies, which are usually few and different, from the data than to build a model to profile normal instances. IFs are based on an ensemble of isolation trees. For isolating anomalous observations, randomly sub-sampled data is processed into a tree structure based on random feature selection until the only data point left in the partition is the observation. Multiple random trees are formed recursively, and data is partitioned in each one of them unless all observations are isolated for each iteration. An observation's path length from the root node to the leaf determines the number of partitions required to isolate it. Deeper-seated samples require more cuts to isolate them, so they are less likely to be anomalies. Likewise, the samples with the shortest average path length indicate anomalies as it was easier for the trees to isolate them [35][36].

Any anomaly detection method requires an anomaly score. IF calculates an anomaly score for an observation x in a data set of n instances as follows:

$$s(x;n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (1)$$

where h(x) is the number of edges in a tree for a point x,

E(h(x)) is the mean of all the h(x) from all isolation trees, and

c(n) is a normalization constant for a dataset of size n.

The anomaly score s is in the [0,1] range, with values closer to 1 being more likely to be anomalous.

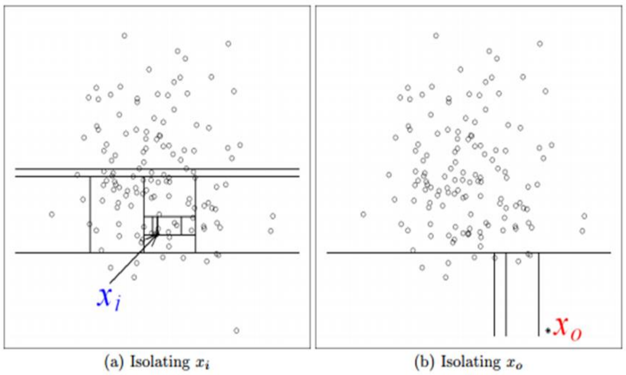

Figure 4 Isolation of a standard observation xi versus an anomalous observation x0 [35]

The number of trees t and the sub-sampling size are important parameters when training an Isolation Forests algorithm. The tree height limit l is calculated according to the sub-sampling size. The sub-sampling process regulates the amount of training data and is a crucial factor because it has a direct impact on the classifier's performance. As a different sub-sample is chosen to build an isolation tree each time, this can aid in more effectively isolating the anomalies. For anomaly predictions from a trained model, equation (1) is used to calculate the anomaly scores, as discussed in [36].

Due to its ability to identify anomalies without relying on any distance or density-based measure, IF is fast and computationally efficient. Moreover, it has a linear time complexity with a low constant and a low memory requirement.

B. AI in Decentralized Marketing

Once a DAO has been launched on a blockchain, the responsibility of marketing it lies with the team that created the DAO in the first place, as well as the members/stakeholders of the DAO. There are several ways by which this is achieved, including DAO marketing agencies (such as Single Grain, GrowthChain, DAO Maker, etc.), social media promotions, airdropping tokens, and so on. In this work, we propose a way to decentralize DAO marketing using incentivization, utilizing the incorporation of AI with smart contract(s) underlying a DAO.

The community awareness and participation of a DAO can be further amplified if we utilize incentivization mechanisms so that even the general public is motivated to act as marketing agents of the DAO. Whenever a person posts something positive about a DAO on social media platforms, the DAO can reward the person with DAO tokens, thereby embodying an essential element of blockchain technology (viz., incentivization). Integration of AI and blockchain can be of assistance here. Smart contract(s) underlying a DAO can utilize an AI model for performing Sentiment Analysis of social media posts (template of which would be specified by the DAO) referencing the DAO in question.

Sentiment analysis (SA) is an important research area with many applications. Sentiment Analysis (SA), also known as Opinion Mining (OM), is the computational examination of people's opinions, attitudes, and feelings toward an entity [37]. To address the Sentiment Analysis (SA) problem, various approaches have been proposed [38], including Lexicon-based techniques(dictionary-based and corpus-based)[39], Machine Learning-based techniques(traditional models and deep learning models)[40], and Hybrid approaches(combination of lexicon- and machine learning-based techniques)[41]. With advancements in Deep Learning, DNNs, CNNs, and hybrid approaches (a combination of several models and techniques) have been identified as the most widely used models for sentiment polarity analysis [42]. An ensemble of Convolutional Neural Networks (CNNs) and Long Short Term Memory (LSTMs) networks with multiple convolution operations has been shown to achieve state-of-the-art results on Twitter Sentiment classification [43]. For pre-training word embeddings, the model uses a significant amount of unlabeled data. Afterward, a subset of the unlabeled data is utilized to fine-tune the embeddings via distant supervision. Ensembling of several CNNs and LSTMs is done to boost performance [43]. Lately, with the huge popularity of language models, such as BERT[44], which is pre-trained on massive amounts of unlabeled data and fine-tuned on downstream tasks, state-of-the-art results have been achieved in a variety of NLP tasks[45]. BERTweet [46] is one such model, which is currently the state-of-the-art tweet analysis model[47]. It is based on the BERT architecture but pre-trained for English tweets, outperforming previous state-of-the-art models on tasks such as Part-of-speech tagging, Named-entity recognition, and text classification [46]. Its pre-training approach is based on RoBERTA[48], which is an optimization of the BERT pre-training procedure for more robust performance.

In our proposed use case, the AI model will generate a sentiment score that will aid in assessing the sentiment of a social media post about the DAO. The score could, for example, be on a scale of 0 to 10, reflecting the most negative to the most positive sentiment. The smart contract can then utilize that in combination with the social media post engagements (likes, shares, comments, etc.) and the reach of the original poster to reward the "marketer" proportionately.

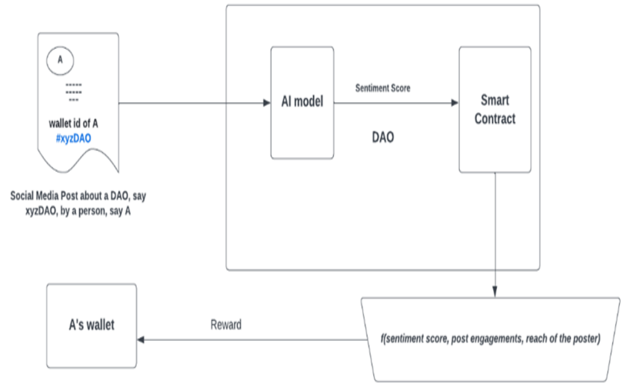

Figure 5 AI assisting in decentralized marketing

C. Fashion Designing in Gaming

Owning a gaming asset/property is a thriving market in today's age. Global gamers spent $54 billion on in-game assets in 2020. It has been estimated In-game sales might reach $74.4 billion by 2025, according to Statistica[49]. These gaming assets include clothes and other fashion entities within the games, likewise in decentraland metaverse robolux. Fashion in gaming imparts beauty and an element of uniqueness without interfering with the skills [50]. This proposed use case generates fashion articles by analyzing the market trends and selling them as NFTs. NFTs are mere digital authentication certificates. These are protected by blockchain technology and are, therefore, tamper-resistant. They are crucial for preserving brand value. This enables fashion firms to sell their items in the virtual world via NFTs, in addition to the actual world. This is particularly prevalent within the gaming community, where player personas attend fashion shows, purchase, and interact with one another [51]. The reason to utilize the services of AI in art generation is AI monitors buyer's preferences and trends in preferences, provides personalized experiences for various customer groupings, and enhances product suggestions [52]. AI is also assisting brands in establishing stronger relationships with their consumers by anticipating their needs. Digital fashion still being in its infancy, and designers and businesses have a vast universe to pursue. Digital apparel is exclusively available for and on social networking platforms. With fashionable fads, including 'fashion hauls' and 'weekly lookbooks' on TikTok and Instagram, in which influencers purchase and don many garments solely for certain videos, digital apparel is unquestionably the more eco-friendly and viable alternative. While it may be difficult to see oneself wearing digital garments, it is an excellent method for marketing promotions that allows for enhanced visualization and less waste.

The way to generate the required fashion articles are GANs. Generative Adversarial Network (GAN) is one of the AI models that is substantive for creating artworks. GANs, utilize computational models for an entirely new intention: generative modeling. Generative modeling is an unsupervised process in machine learning that entails systematically detecting and learning regularities or features in incoming data so that the model is used to produce or output new instances which reasonably may be chosen from the given dataset [53]. The group of generative models includes Generative Adversarial Networks. These models have the capacity to generate fresh data out of their own. GANs are composed of two neural networks, competing against one another, the Generator and the Discriminator. They act like a Fraudster and an inspector, where Generator always makes bogus data (the artworks in our example) and the discriminator is continually attempting to capture the Fraudster. According to a brief description of Generator's role, the discriminator will be fooled by a dataset that seems to be an actual artwork. In addition, the discriminator's role is to distinguish between genuine and phony art. Until it can trick the discriminator, the generator will keep increasing its output. To put it another way, this indicates that the generator has figured out how to mimic the underlying distribution in dataset in order to produce similarly realistic paintings. Training of both networks occur in Tandem, the first Generator is trained for few epochs then the discriminator. Mathematically, the objective function of a GAN can be expressed as:

$$E_x[\log(D(x))] + E_z[\log\left(1 - D\left(G(z)\right)\right)] \tag{2}$$

Here,

D(x) is the discriminator's estimated probability that real data instance x is real.

Ex represents the expected value over all real data instances.

G(z) indicates the generator's output when given noise z.

D(G(z)) is the discriminator's estimated probability that a fake instance is real.

Ez represents the expected value over all the random inputs to the generator, i.e., the expected value over all generated fake instances G(z)[54].

The generator aims to minimize the objective function while the discriminator aims to maximize it.

Several GANs that could find application for this use case include Deep Convolutional Generative Adversarial Networks (DCGANs)[55][56], Wasserstein GAN with Gradient Penalty (WGAN-GP)[57], StyleGAN[58], StyleCLIP[59], and Text2Mesh[60], DRAGAN[61][62], CESAGAN[63], 3D-GAN[64], and a newer technique that is built upon it, known as Inverse Graphics GAN[65][66].

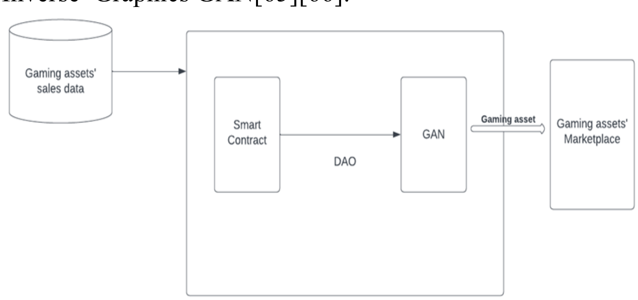

Figure 6 AI generating gaming assets based on trends for Blockchain-based gaming

D. Real Estate

Real estate events are typically handled offsite, including face-to-face meetings with numerous organizations. Blockchain, nevertheless, opens up options to modify this. The decentralized, secure nature of the blockchain makes it ideal for recording and transferring possession of assets. This has the capability to be a game-changer for the real estate market.

The emergence of smart contracts on blockchain platforms now enables commodities, including real estate, to be tokenized and traded, such as cryptocurrencies [67]. Moreover, the use of artificial intelligence in real estate may also help organizations determine the ideal timing to buy or sell real estate, as well as predict future lease or selling prices [68]. AI forecasts the future value of a region based on the town's geographic location and the potential for future developments or physical effects. In addition, it may estimate which neighborhood is more likely to be the target of a burglary or other crime because this is an essential consideration when purchasing a real estate. In our use case we are utilizing the services of AI to provide robust recommendations, price prediction and disaster management, and smart contracts to tokenize and transact the estate. When specific requirements are satisfied, a smart contract is utilized to take possession of the asset to the buyer. Using them, one may check the validity of many layers of information, as well as transfer rights from one person or organization to another in a more effective way.

IV. Experimentation and Results

We simulated a simple blockchain that runs smart contracts. The modules we designed are Block, Blockchain, Smart Contract, and Data. The Block consists of information about the timestamp of its genesis and hashes of data and the previous Block. The Blockchain module runs the smart contract and passes the output of the state attained by running the contract to the Block module, creating a new block that stores the newly acquired state. It is here where the decision of how many blocks need to be added is made. The Smart contract module has functions that add or subtract a certain number of tokens to the account when invoked. The decision is taken involving an AI-based anomaly detection model(anomaly_model.py). Figure 7 presents the simulation of the anomaly detection model in a smart contract.

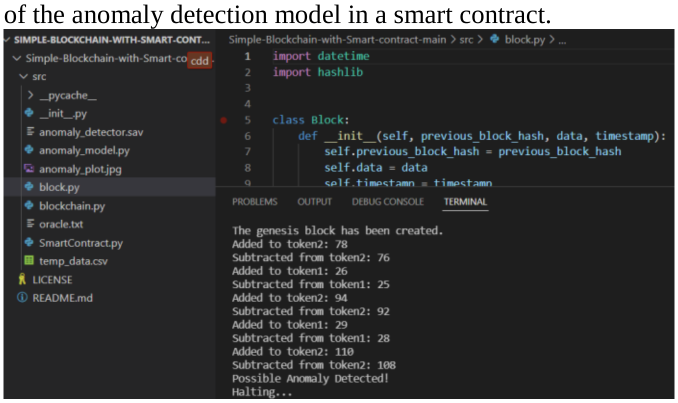

Figure 7 Simulation of anomaly detection model in a smart contract

We use the scikit-learn library's implementation of the Isolation forest model (sklearn.ensemble.IsolationForest) to train an anomaly detector on a small sample dataset(temp_data.csv). We use a simple text file as a substitute for an oracle(oracle.txt), supplying a series of values to the smart contract, which would be detected for anomalies by our anomaly detector. In case there is a detected anomaly, no operation is performed on the tokens of the Data object, as displayed in the graph (Figure 8). Otherwise, the smart contract operates normally, adding or subtracting values to the tokens.

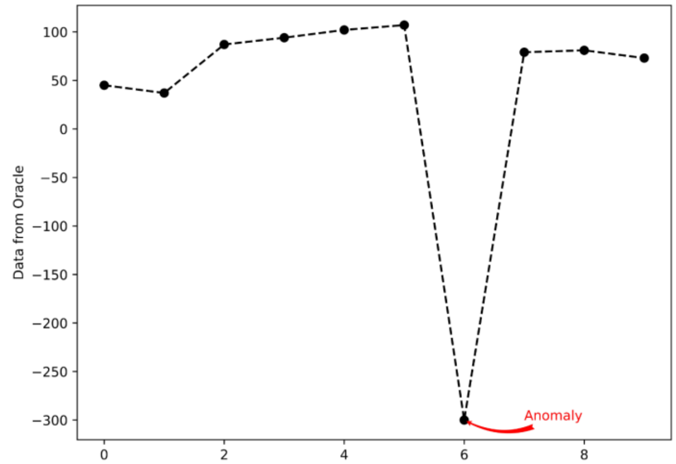

Figure 8 Graph demonstrating anomaly detection in oracles

V. Challenges in Conjunction of AI and Blockchain

The blockchain artificially intelligent conundrum relies heavily on the ability of AI to provide innovative solutions with cognitive characteristics, as the basic purpose of artificial intelligence (AI) is to reduce human mistakes and speed up procedures. Thus, it is evident that both AI and Blockchain are aimed at speeding up procedures. Employing each of these simultaneously has the potential to increase the use of blockchain in a wide range of industries [69]. To determine whether or not AI and Blockchain are complementary technologies, it is necessary, however, to investigate the distinctions that exist between the two. Centralization is the primary distinction that can be made between artificial intelligence and blockchain technology. Artificial intelligence is fundamentally a centralized technology, while blockchain is constructed on the principle of decentralization. The second significant distinction amongst two is that of transparency. Blockchain is simple to comprehend and provides greater prospects to commence, whereas artificial intelligence is comprised of numerous mathematical computations and notions. These differences result in to certain challenges that are required to be dealt with while converging two technologies. Challenges include.

Potential sophisticated answers may become fuzzier in AI and Blockchain collaboration[70].

As Blockchain-specific hardware that is suitable for mining and decentralized storage is still under research, hence AI and blockchain combo lacks proper infrastructure.

Regulated access control of data, in order to preserve privacy of public blockchain data, imposes restrictions on the prediction and analysis of ML models.

Since there are not any defined standards for blockchain, it really is crucial to keep a watch on the expense of smart contracts deployment, transaction costs, and on-chain or off-chain storage.

The current execution of smart contracts and VMs is handled by CPUs, although owing to the complexity of AI systems, no public blockchain has the processing capacity to successfully run AI algorithms. In their current iteration, these solutions only allow businesses to do AI tasks off-chain or on a side-chain, before transferring the findings back to the main chain.

VI. Conclusion

The focus of our paper is on incorporating AI in smart contracts that is missing in the existing state-of-the-art frameworks. A brief overview of existing AI-based blockchain platforms is presented. In this paper, novel use cases are proposed in which AI is embedded in smart contracts to incorporate AI characteristics in the blockchain. The essential premise of the first use case is to build a model to characterize and isolate anomalies from the data, which are often rare and distinct. In the second use case, we've outlined, the AI model will produce a sentiment score that will help determine the sentiment of readers about DAO. By examining market trends, another use case proposal creates fashion items that are then sold as NFTs. In our last use case, we are using smart contracts to tokenize and trade the estate and AI to give strong recommendations, price prediction, and catastrophe management.

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